

Wookje Han

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SUMMARY

AI performance engineer working on LLMs, video models, and recommender systems across the stack, from inference engines to GPU kernels. Focused on GPU kernel optimization for low-latency inference and high-performance training on recent hardware architectures.

TECHNICAL SKILLS

Languages: Python, C/C++, CUDA C++, PTX

Frameworks, Libraries: CUTLASS, CuTeDSL, NCCL, vLLM, SGLang, FlashInfer, PyTorch, OpenAI Triton

Developer Tools: Docker, AWS, GCP, Nsight Systems, Nsight Compute

EDUCATION

Columbia University

MS in Computer Science

New York, NY

Sep 2023 - Dec 2024

Seoul National University (SNU)

BS in Computer Science and Engineering, Graduated Summa cum laude

Seoul, Korea

Mar 2017 - Aug 2023

EXPERIENCE

Senior AI Technology Developer

NVIDIA

Mar 2026 - Present

Santa Clara, CA

- Collaborated with a partner company to optimize end-to-end inference for its LLM on vLLM and SGLang.
- Implemented custom attention kernels for Hopper, Blackwell, and Rubin across diverse attention masks and data types, achieving state-of-the-art training and inference performance.
- Developed high-performance GEMM on recent architectures for low-latency and high-throughput workloads.
- Optimized fused multi-GPU communication and computation kernels using NVSwitch for end-to-end inference.
- Contributed a custom GB300 kernel to [NVIDIA's MLPerf Inference v6.0 submission](#).

AI Technology Developer

Feb 2025 - Feb 2026

- Optimized custom kernels and end-to-end deep learning workloads (e.g., recommender systems, LLMs, and DiTs) on recent architectures using CUDA, CUTLASS, CuTeDSL, and OpenAI Triton.
- Led a partner company's adoption of Blackwell for training by implementing custom attention forward and backward kernels in CUTLASS, achieving 4X over the OpenAI Triton kernel.
- Optimized high-performance Group GEMM forward and backward kernels on Hopper, 3.8X faster than OpenAI Triton across both low-latency and high-throughput regimes.
- Delivered technical lectures to partner companies on (i) CUTLASS and CuTeDSL, (ii) writing high-performance GEMM kernels, and (iii) recent architecture-specific optimization.

AI Technology Developer Intern

May 2024 - Aug 2024

- Implemented a CUDA kernel for the encoder layer of a Generative Recommendation model, utilizing Hopper Architecture features (TMA and WGMMMA), resulting in a 2X performance improvement over the Triton kernel.
- Built MMA and epilogue fusion using CUTLASS, accelerating the Generative Recommendation Model by 1.16X.
- Analyzed the characteristics and detailed statistics of bottleneck kernels in the Generative Recommendation Model using Nsight Systems and Nsight Compute.

Software Systems Research Intern

Software Platform Lab, SNU

Dec 2021 - Feb 2023

Seoul, Korea

- Constructed a tool to distribute language model's memory load and computation to multiple GPUs with PyTorch.
- Established multi-node environment using docker on GCP for efficient memory management of DL training.

PROJECTS

LLVM Compiler Optimization | C++, LLVM Compiler, Git, Docker, Linux

Mar 2022 - Jul 2022

- Led a team of 4, devising optimizations for LLVM compiler with C++ and reduced 42% of cost for a given machine.
- Built automated unit tests utilizing CMake and GitHub Actions for continuous integration.

ArchPresser | Python, NumPy, pybind, Git, Linux

Sep 2021 - Dec 2021

- Constructed a program to generate a panoramic dental x-ray image from 3D CT scans with Python and C++.
- Analyzed bottleneck and used pybind to integrate C++ code with Python, leading to 20x faster application.